INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

BIOLOGY **9648/03**

Paper 3 Applications Paper and Planning Question **11 September 2017**

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**16**
2	**11**
3	**13**
4	**12**
5	**20**
Total	**72**

This document consists of **14** printed pages.

Answer **all** questions.

1 Erythropoietin is a glycoprotein hormone which plays an important role in erythropoiesis (production of erythrocytes) from haematopoietic stem cells. It is a cytokine that binds erythrocyte precursors in the bone marrow.

(a) Describe how red blood cells are formed from haematopoietic stem cells.

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.... [2]

The erythropoietin gene can be cloned into bacteria host cell for production of the protein using genetic engineering. Erythropoietin is administered to anaemic patients who are unable to produce sufficient erythrocytes. Fig. 1.1 shows the human erythropoietin gene ligated with *Bam*HI and *Hin*dIII linkers.

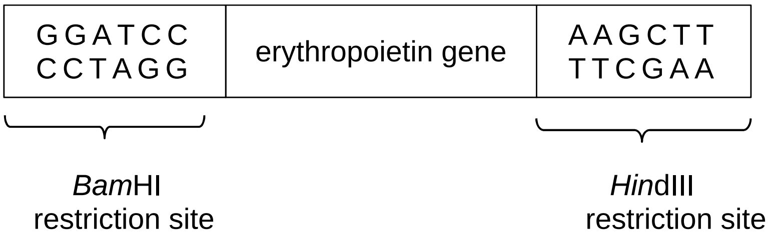

Fig. 1.1

(b) *Bam*HI cuts between guanine nucleotides while *Hin*dIII cuts between adenine and guanine nucleotides.

(i) Illustrate the resultant restriction fragment from Fig. 1.1 after restriction digestion by *Bam*HI and *Hin*dIII.

[2]

(ii) Suggest **two** reasons why the erythropoietin gene is cut using two different restriction enzymes.

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.... [2]

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(iii) Explain why the use of restriction enzymes like *Bam*HI has a higher chance of producing recombinant DNA with plasmid compared to *Sma*I which cuts symmetrically at its restriction site.

[2]

(c) Both *Hin*dIII and *Bam*HI are restriction enzymes commonly found in bacteria.

(i) Describe the natural function of restriction enzymes.

[2]

(ii) Explain why the bacterium's own DNA is not digested by its own restriction enzymes.

[2]

(d) **(i)** The human erythropoietin gene was isolated from the genomic library and introduced into a bacterium host cell. The resultant erythropoietin protein produced was non-functional and found to be longer than normal.

Explain why the erythropoietin protein produced by bacteria was longer than normal.

[2]

(ii) Another copy of the erythropoietin gene was isolated from cDNA library. Subsequent introduction of this erythropoietin gene into *E. coli* produced erythropoietin of the correct length but is still non-functional.

Suggest why this is so.

[2]

[Total: 16]

2 Read the following passage.

Can't go for liver transplant? There's hope still.
(Excerpt from The Straits Times, 12 Jul 2017)

A transplant can cure end-stage liver cirrhosis - hardening of the liver - but not all patients have this option. For those who are not eligible for a transplant, an alternative may be in sight.

A clinical trial was launched yesterday to explore the use of stem cells to reverse liver cirrhosis. In the study conducted by a multi-centre team led by the National University Hospital (NUH), doctors aim to determine if stem-cell therapy can improve liver function. Stem cells will be taken from a patient's own bone marrow and will be isolated and injected directly into the patient's liver to initiate the repair.

The $2.6 million Phase III trial will use biopsy and clinical measurements of liver function to test the efficacy, effectiveness and safety of the stem-cell treatment. The study is funded by the National Medical Research Council, and a total of 46 patients will be recruited.

Liver cirrhosis is caused by diseases such as chronic hepatitis B and non-alcoholic fatty liver disease. A liver transplant provides a definitive cure to end-stage cirrhosis. However, in Singapore, less than 5% of end-stage liver cirrhosis patients receive a liver transplant.

The number of people on the waiting list for a liver transplant has been increasing over the years, according to statistics from the Ministry of Health. Last year, there were 57 on the waiting list, up from 9 in 2007. There are around 50 waiting for a liver transplant this year.

While similar therapy treatments have been conducted overseas in countries such as Egypt and India, they have not been fully evaluated for efficacy.

Associate Professor Dan Yock Young, a senior consultant in the division of gastroenterology and hepatology at NUH, said: "We are conducting the study in a systematic and scientific manner in order to get definitive evidence of the effects of the treatment."

He said stem-cell therapy is not a substitute for a liver transplant but provides an option for those who are not eligible for one.

(a) With reference to the features of stem cells, explain how stem-cell therapy can improve liver function in patients (line 6).

[3]

(b) Explain why stem cells are taken

(i) from the bone marrow (line 7);

[2]

(ii) from the patients themselves (line 7).

[1]

(c) Suggest why less than 5% of end-stage liver cirrhosis patients receive a liver transplant (line 15).

[1]

(d) Suggest why stem-cell therapy is not a substitute for a liver transplant (line 25).

[1]

Haemochromatosis is an autosomal recessive disease that leads to excessive iron absorption from diet leading to abnormal deposition in liver and other organs. The iron built up can damage liver cells leading to liver cirrhosis.

The HFE gene at locus 6p22.2 undergoes C282Y mutation that results in 2 instead of 1 *Rsa*1 recognition site.

(e) A DNA fragment was extracted from a HFE carrier, subjected to polymerase chain reaction (PCR), followed by *Rsa*1 restriction digestion and separated using gel electrophoresis. The resultant gel is shown in Fig. 2.1.

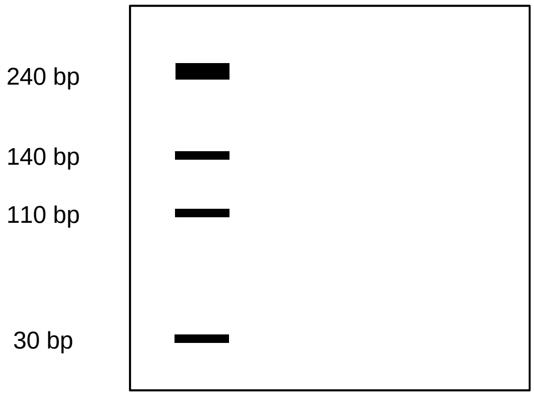

Fig. 2.1

(i) Illustrate the restriction map for both the normal and mutant HFE alleles.

[2]

(ii) Calculate the number of PCR cycles needed to obtain 1 million copies of the HFE allele from one copy.

[1]

[Total: 11]

3 Cultivated rice, *Oryza sativa*, is often grown in fields flooded with water.

Some varieties of cultivated rice are able to grow long internodes (a length of stem between leaves) when they are submerged in water, keeping the leaves and flowers above water level. These varieties are known as deepwater rice. The snorkel genes SK1 and SK2, thought to be responsible for this response, were identified in a variety of deepwater rice, C9285. A non-deepwater variety, T65, did not have these genes.

The expression of SK1 and SK2 genes result in increased production of gibberellin, GA. GA is a plant growth regulator which promotes cell growth and elongation. Fig. 3.1 shows the effect of submergence on GA production in C9285 and in T65.

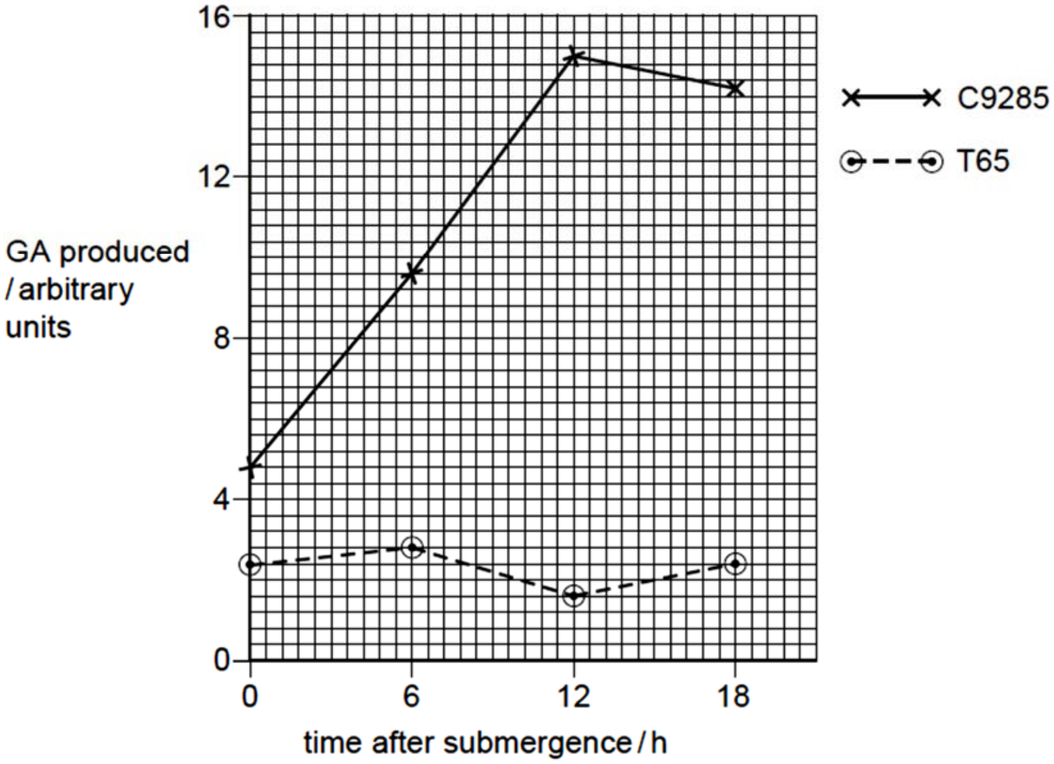

Fig. 3.1

(a) Compare the effect of submergence on GA production in C9285 and T65.

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.... [2]

The effect of duration of submergence of rice plants was also investigated. Fig. 3.2 shows the result on internode elongation.

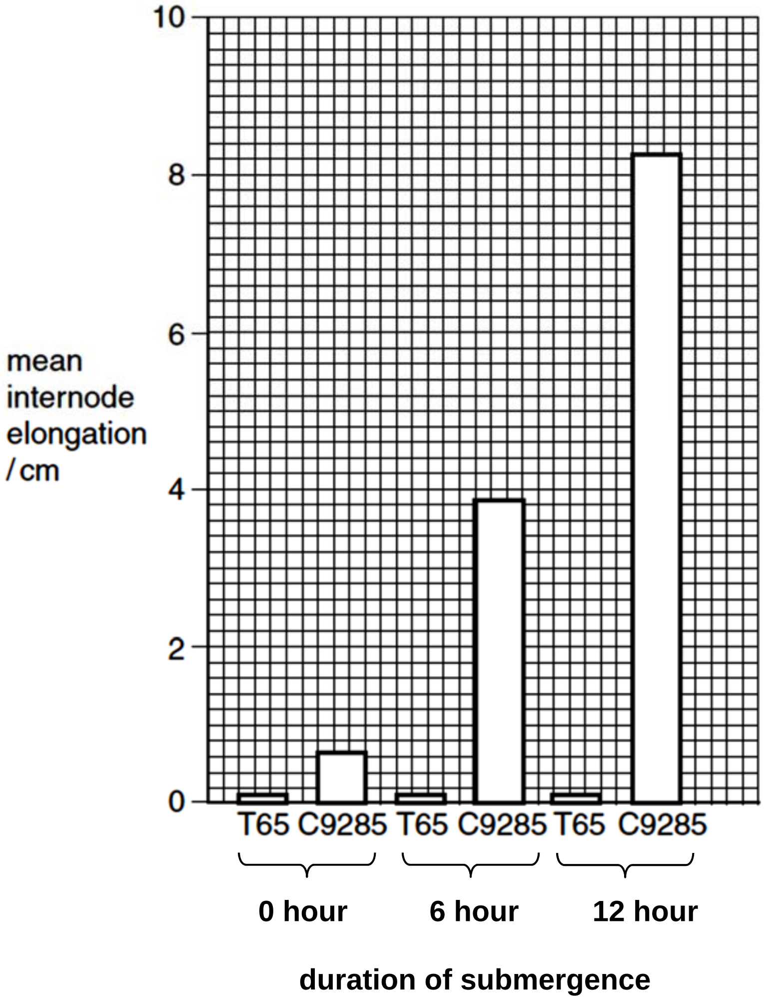

Fig. 3.2

(b) With reference to Fig. 3.2 and the information given, explain the effect of submergence on internode elongation in T65 and C9285.

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.... [2]

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(c) Deepwater rice is the main food crop in many parts of the world that undergo flooding in the rainy season. However, many varieties of deepwater rice have lower yields than non-deepwater varieties. Deepwater rice variety with high yield can be produced using genetic engineering technology. However, farmers attempted to produce such variety using artificial selection instead.

(i) Describe how cells from high yield rice plant can be genetically modified to have deepwater characteristics.

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(ii) Describe how genetically modified rice plant cells in **(c)(i)** can be grown into rice plants.

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(iii) Explain why farmers used artificial selection rather than genetic engineering to produce deepwater rice variety with high yield.

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[Total: 13]

Planning Question

Most mineral elements are essential for the healthy growth of plants. However, lead is known to be toxic and impede plant growth. Lead forms strong covalent bonds with a plant enzyme and its binding alters the 3D conformation of the enzyme's active site.

You are to design an experiment to investigate the effect of different concentrations of heavy metal lead (as lead chloride solution) on the growth of cress seedlings.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 50 cress seedlings
- Petri-dish with cotton wool
- 2.0 mol dm^{-3} lead chloride solution
- distilled water
- glass rod
- pH buffer (pH 6)
- ruler
- lamp
- weighing balance
- a variety of different sized beakers, measuring cylinders, syringes and pipettes for measuring volumes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:
- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections (a), (b) etc., as indicated in the question.

(a) Explain the benefits of the human genome project. [8]

(b) With reference to genetic diseases, explain gene therapy treatment, using both viral and non-viral delivery systems. [6]

(c) Discuss the social implications of genetically modified crop plants. [6]

[Total: 20]